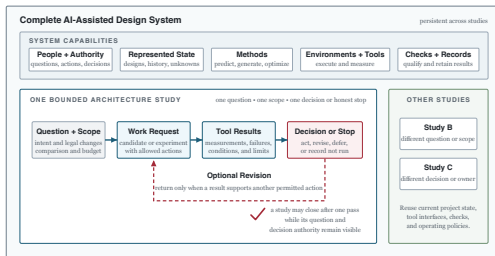


Architecture 2.0

Principles for AI-Assisted System and Chip Design



Three fields, one architecture decision

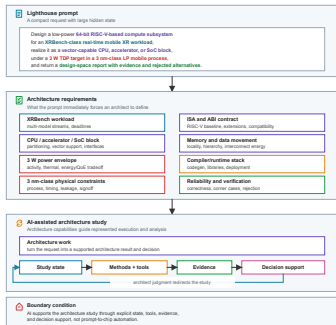
The useful intersection is not “AI for chips.” It is a supported architecture decision.



- Architecture supplies the tradeoff and system boundary.
- ML supplies possible prediction, generation, and optimization methods.
- EDA supplies executable constraints and physical observations.

The one-sentence trap

A compact product request hides a full-stack architecture program.



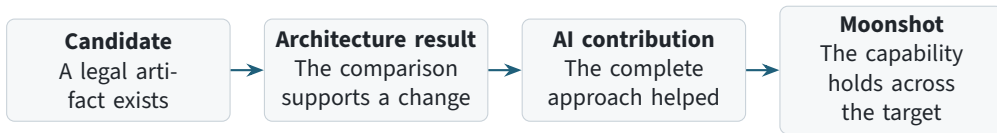
Lighthouse target

- low-power, 64-bit RISC-V compute subsystem
- real-time mobile XR workloads
- software, RTL, physical, verification, and product constraints

Source: Architecture 2.0, Chapter 1; XRBench and ILLIXR ground the workload class.

Four different success claims

Output, architecture value, AI contribution, and moonshot completion are not interchangeable.



Each arrow needs a separate test.

The Architecture 2.0 bet

Improve the architecture result, reduce the total cost of reaching a comparable result, or both.

Architecture quality

- performance, power, area
- reliability, security
- software and product fit

Total cost

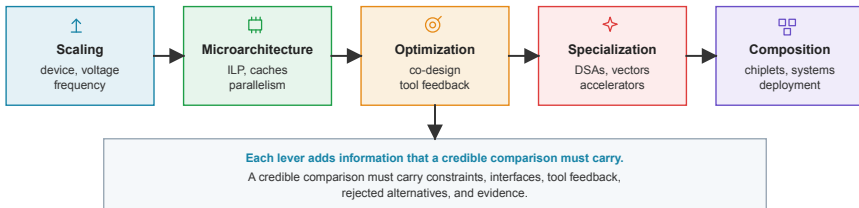
- model and tool calls
- queue and license time
- human setup, review, repair

Failure behavior

- invalid candidates
- missed constraints
- unsafe or uncontrolled actions

Every efficiency lever adds obligations

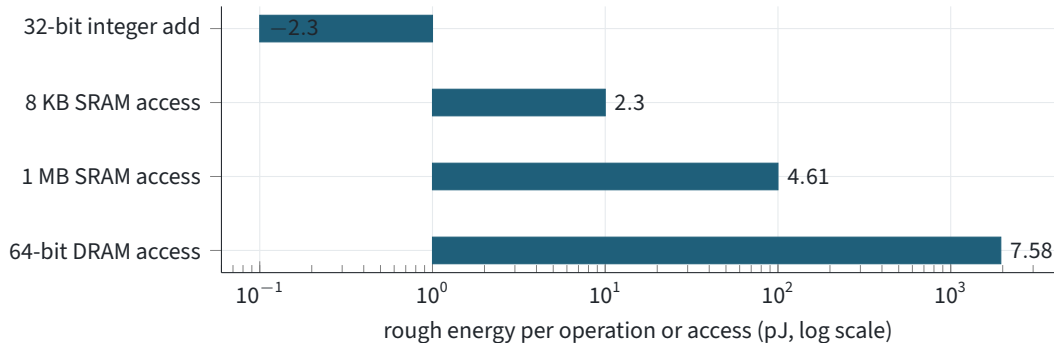
Scaling gave way to specialization, composition, and cross-layer work.



Source: Dennard scaling, Amdahl's law, dark silicon, chiplets, and warehouse-scale computing are established architecture results summarized in Chapter 2.

Data movement can dwarf arithmetic

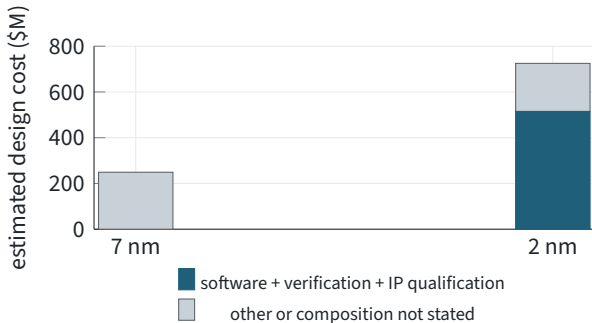
The system boundary decides whether a local compute gain matters.



Source: Horowitz, ISSCC 2014 rough 45 nm energy anchors. DRAM is plotted at the midpoint of the reported 1300–2600 pJ range; these are order-of-magnitude anchors, not current-node predictions.

Cost has moved beyond drawing the RTL

At advanced nodes, software, verification, and IP qualification can dominate estimated design cost.



71%

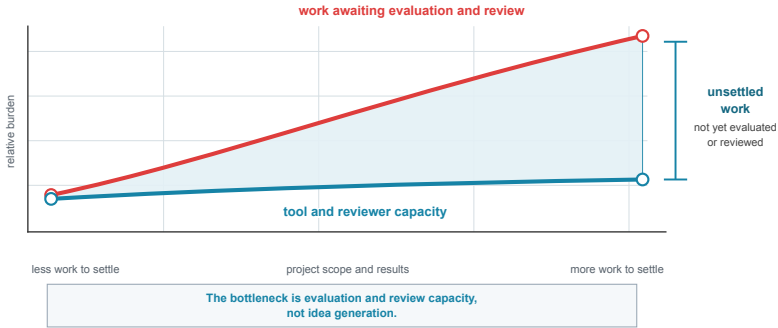
of the \$725M 2 nm estimate was attributed to the combined software, verification, and IP-qualification categories.

The 7 nm composition was not published, so the chart does not invent it.

Source: Arm Holdings, Form 424B4 (2023), reporting July 2022 IBS estimates: \$249M at 7 nm and \$725M at 2 nm.

Production can outpace examination

Faster candidate creation is useful only if evaluation and review can keep up.



Diagnose the limiting work first

Different delays call for different interventions.

Missing information	retrieve, reconcile, or collect the state
----------------------------	---

Invalid or unhelpful choices	constrain or change the candidate space
-------------------------------------	---

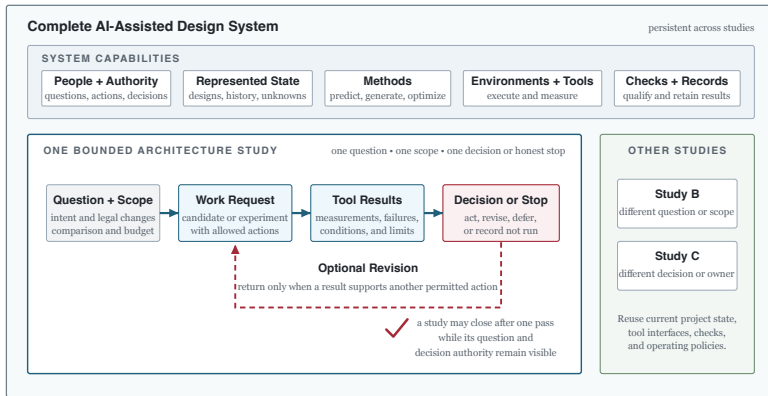
Expensive evaluation	predict, screen, or allocate stronger checks
-----------------------------	--

Overloaded review	change capacity, routing, or review preparation
--------------------------	---

Unclear decision	name the action, owner, and evidence threshold
-------------------------	--

“Use AI” is not a diagnosis.

The object being engineered is the complete system



System, study, and loop are different

The loop is a possible behavior inside a study, not the whole thesis.

Complete system

- persistent people, methods, tools, state, checks
- supports many studies

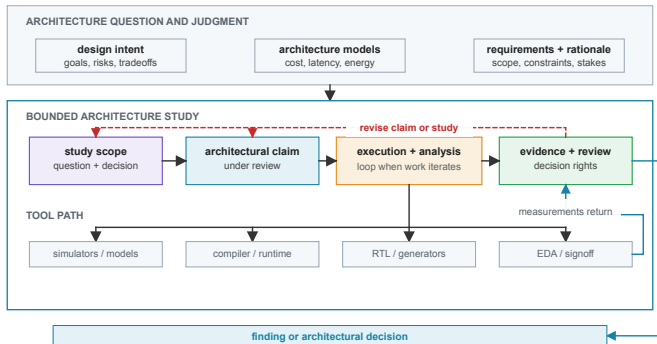
Bounded study

- one question and scope
- one decision or honest stop

Optional loop

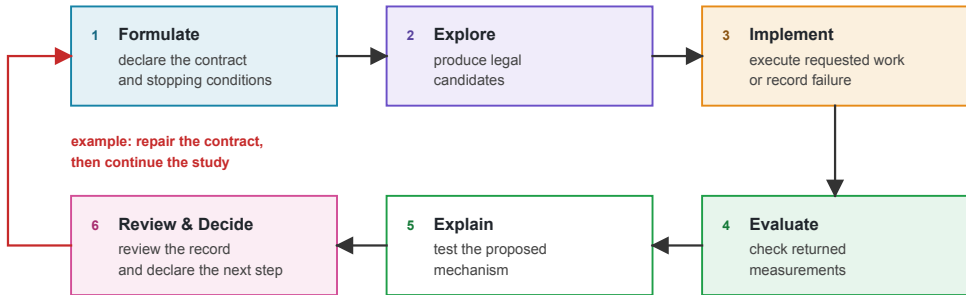
- another permitted action
- only when the result supports it

A bounded study connects question to decision



Six distinct kinds of work

The stages are separated because they answer different questions and fail differently.



Repair the part that failed

Bad question

reformulate

Weak comparison

change alternatives or budget

Broken execution

repair the environment

Inadequate check

strengthen measurement

Failed explanation

narrow or retest mechanism

Unresolved decision

defer, reject, or assign work

Iteration should have a technical reason and a destination.

One record holds the comparison together

Another architect should be able to reconstruct what changed, what ran, what failed, and what the result supports.



The record connects the technical argument; it is not the argument itself.

The project knows three different things

General knowledge mechanisms, specifications, papers, slow-moving relationships

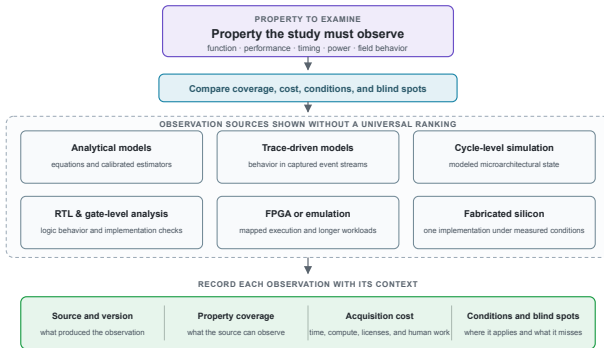
Current project state exact design, workload, software, constraints, owners

Project history runs, failures, rejections, adoption, changed conditions

A model's general knowledge is not live project state.

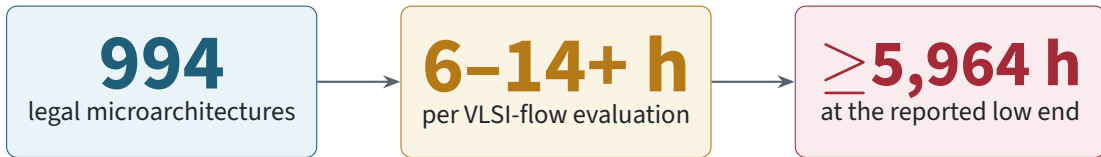
Observation sources are not interchangeable

Choose the source by the property the study must observe, not by a single fidelity ranking.



An architecture sample can be expensive

Data acquisition is part of the architecture method, not background plumbing.

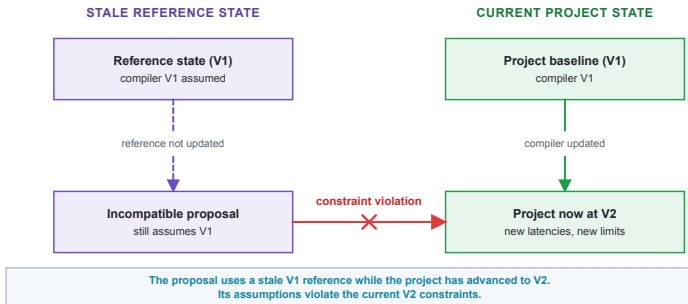


Ten analytical calls, ten cycle simulations, and ten physical-design runs are not ten equivalent samples.

Source: BOOM-Explorer: Bai et al., ISCA 2021. The final value is a lower-bound multiplication of the reported sample count and low-end runtime.

Representation determines reachable designs

A legal proposal can still be invalid when project state and tool-facing artifacts disagree.



Role comes before method

Do not collapse an engineering job, a method family, and an implementation into “the AI.”

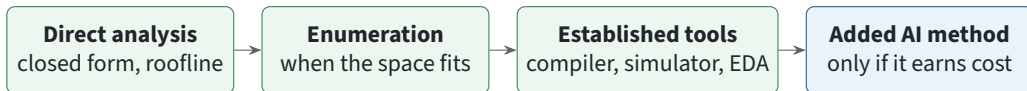
1. Engineering role propose, estimate, select, check, operate, interpret

2. Need for an added method direct tools or existing analysis may already suffice

3. Method family generation, prediction, optimization, or a combination

4. Implementation classical, learned, language-model based, or hybrid

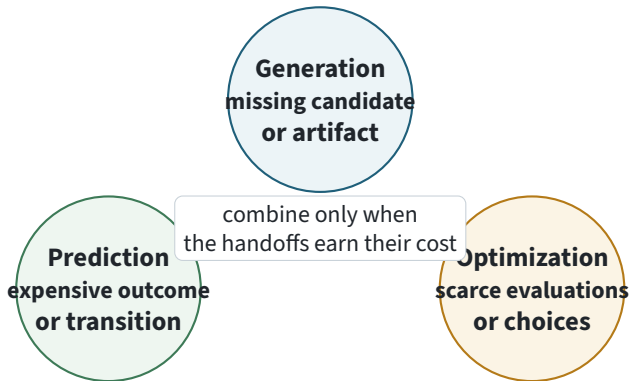
Use the smallest sufficient approach



**“No added AI” is a valid
architecture-method decision.**

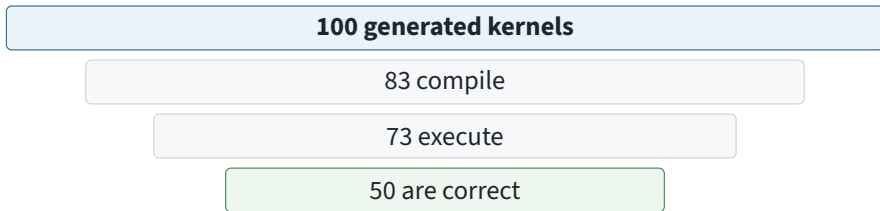
Three method families

Start from what is missing, not from a fixed prediction-generation-optimization sequence.



Generation expands the candidate set

Attempted, executable, correct, and useful are different conditions.

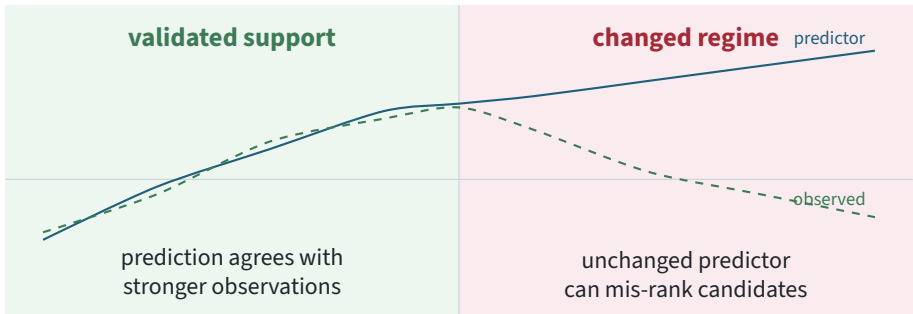


Correct and faster than the baseline is narrower again.

Source: KernelBench, Ouyang et al. (2025), separate 100-kernel error-analysis sample. Per-model one-shot results in the paper show the same attempted → correct → correct-and-faster narrowing.

Prediction trades cost for a support assumption

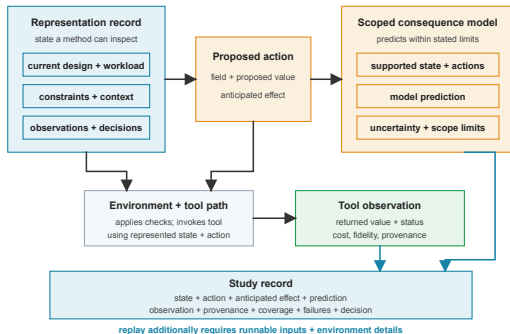
A cheap predictor helps only where its error is acceptable for the decision it screens.



Source: Constructed teaching illustration; Chapter 5 develops calibration and support limits.

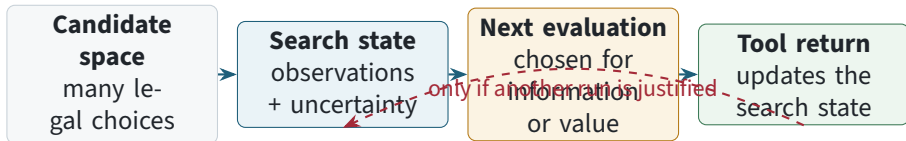
A world model is a scoped predictor

It predicts consequences of state and action; it is not a fourth method family or authoritative project state.

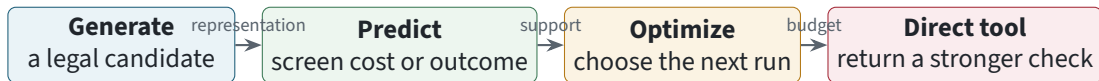


Optimization allocates scarce evaluations

When feedback is expensive, the next evaluation matters more than another unmeasured candidate.



Method handoffs must earn their cost



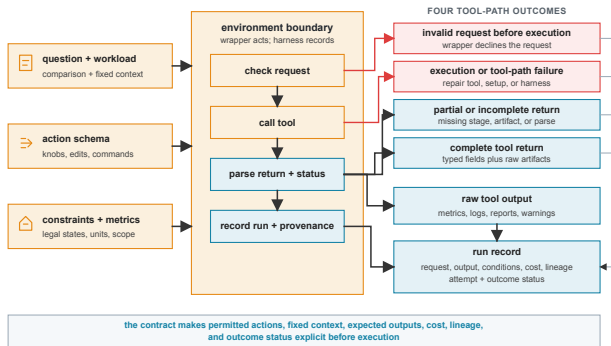
Justified

- each stage removes more cost than it adds
- rejected candidates and failures stay visible

Not justified

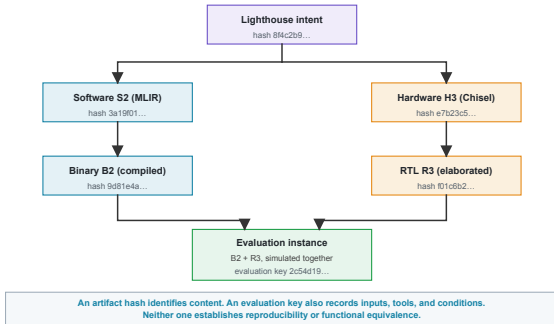
- more components without a named bottleneck
- a cheap proxy replaces the decisive check

An environment is more than a simulator



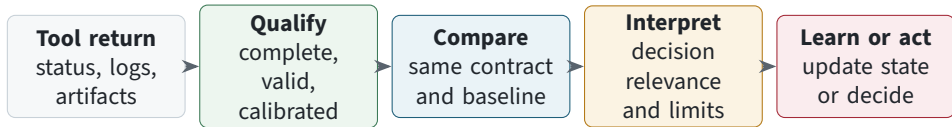
Identity must survive every tool

A successful command is not enough to reconstruct a result.



A return is not yet a measurement

Qualification, comparison, and interpretation give a tool return its meaning.



If qualification fails, return to the tool or environment. Do not report a measurement.

Formal and simulation answer different questions

Formal

- proves or falsifies a stated property
- under a model and assumptions
- may be inconclusive or time out

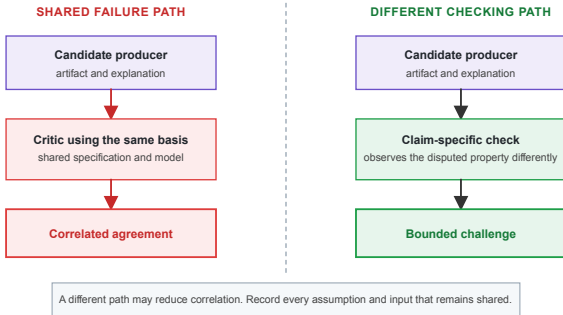
Simulation

- observes exercised behavior
- under selected stimuli and configuration
- cannot exhaust a modern design

Neither check repairs an incomplete requirement or system boundary.

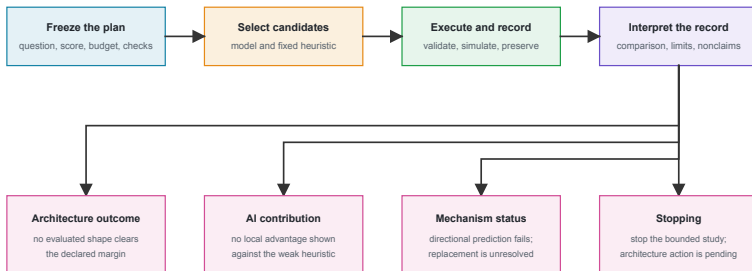
Another model is not an independent check

Independence comes from a materially different observation path, not another fluent answer.



One complete study must resolve four judgments

Architecture outcome, AI contribution, mechanism, and action remain separate.



Equal simulator budgets produced a tie

Neither proposal arm found a nonbaseline shape that cleared the predeclared 1% margin.

Model arm	4 simulator calls	best score 33,782	$32 \times 32 = 16 \times 64$
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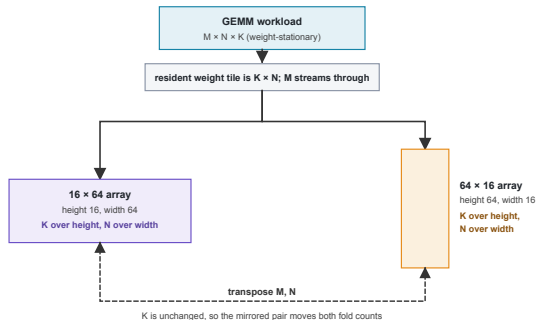
Fixed heuristic	4 simulator calls	best score 33,782	$32 \times 32 = 16 \times 64$
------------------------	-------------------	-------------------	-------------------------------

Architecture outcome: retain the baseline for this bounded study

Source: Retained SCALE-Sim 3.0.0 study in Chapter 8. The score is local and lower is better; none of the values are silicon measurements.

The failed explanation is also a result

The score can remain valid for the local comparison while the proposed mechanism fails.



Prediction

Transposing M and N would reverse the winner.

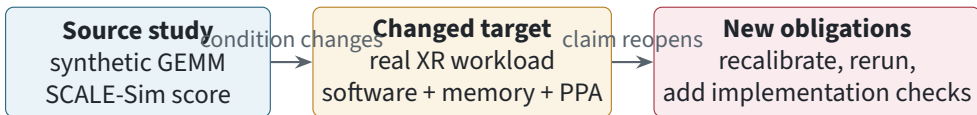
Observed

16x64 still wins the matched contrast.

Keep the outcome. Reject the mechanism claim.

A new decision reopens the claim

Study structure may transfer; a local conclusion does not transfer automatically.



Workload, software, hardware, tools, process assumptions, and decision scope are part of the result.

Evaluate matched complete systems

A model score cannot establish that the complete AI-assisted approach improved architecture work.

Same architecture question, scope, budget, tools, and decisive checks

AI-assisted complete system

- proposed candidates and method calls
- tool routing, retries, and review
- all failures and resource costs retained

Strong comparison system

- established heuristic or human process
- same tool conditions and work scope
- all failures and resource costs retained

Compare architecture outcomes and total resources; report the AI contribution separately.

Metrics people forget to count

Architecture quality and workflow cost need separate, explicit measurements.

Architecture	performance, power, area, reliability, software and product constraints
---------------------	---

Work performed	candidates, model calls, tool calls by type and fidelity, unique simulations
-----------------------	--

Resources	wall time, queue time, compute, memory, licenses, acquired samples
------------------	--

People	setup, review, diagnosis, repair, exception handling, decision preparation
---------------	--

Failures and control	invalid actions, retries, timeouts, recovery, budget violations, unsafe changes
-----------------------------	---

Ten tool calls do not have one common cost.

Red team both the artifact and the workflow

A correct-looking final design can still come from an unsafe or misleading process.

Artifact attacks

hidden logic, backdoors, unsafe interfaces, physical side effects

Evaluator attacks

weakened tests, proxy gaming, omitted conditions, contaminated data

Workflow attacks

silent retries, hidden failures, uncontrolled tool or model spending

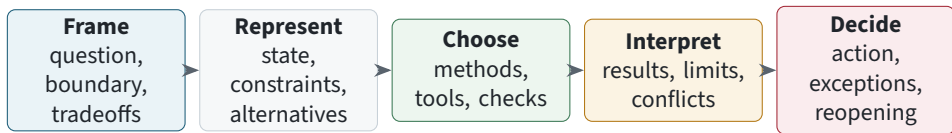
Authority attacks

changed requirements, unauthorized waivers, recommendation treated as approval

Source: Chapter 10; the Malicious LUT research construction shows why checks must reach the final artifact and tool stage, not only the source RTL.

What remains architectural

The architect's role is a positive technical contribution, not whatever automation has not absorbed.



Specialists supply capabilities. The architect integrates them into a supported decision.

Three separations to remember

A candidate is not an architecture result.

A tool return is not yet a measurement.

A favorable assessment is not permission to commit.

Architecture 2.0 turns an ambitious prompt into a supported action. It preserves the system boundary, the limits of each result, and responsibility.